

NAVITH NETHMIKA FERNANDO

SOFTWARE ENGINEERING STUDENT | FULL STACK DEVELOPER

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Panadura, Sri Lanka



ABOUT ME

Highly motivated Software Engineering Student and Full Stack Developer with a strong foundation in Java, Spring Boot, Flutter, and modern cloud technologies. Experienced in architecting scalable backend systems, designing custom data structures, and deploying cloud-native applications. Eager to apply my problem-solving skills to real-world challenges, contribute to innovative projects, and drive technical excellence within a dynamic team environment.

TECHNICAL SKILLS

- Programming Languages: Java, JavaScript, Dart, C/C++, Python, C#, PHP
- Web Technologies: HTML5, CSS3
- Frameworks & Libraries: Spring Boot, Spring MVC, React, Flutter
- Cloud, DevOps & Tools: Docker, AWS, Git/GitHub, Postman
- Version Control: Git, GitHub

PROJECTS

Smart Bus Transit & Ticketing System (LankaTransit)

Engineered a comprehensive, multi-role transit management platform connecting passengers, drivers, conductors, and bus owners.

- Implemented real-time GPS bus tracking and a secure, QR-code-based ticket validation system via a custom mobile scanner.
- Developed role-specific mobile dashboards, including a comprehensive fleet and revenue management module for bus owners.
- Secured backend communications using Spring Security with JWT authentication for seamless and secure Flutter mobile integration.
- **Technologies Used:** Java, Spring Boot, Spring Data JPA, Flutter (Dart), PostgreSQL, REST APIs, Geolocator.

Car Rental System (Microservices)

- Developed a full-stack car rental platform using a microservices architecture.
- Built and maintained backend services with Spring Boot for vehicle inventory, user authentication, and booking management.
- Created a responsive user interface using React.js for browsing vehicles and making reservations.
- **Technologies Used:** Spring Boot, React.js, Microservices, REST APIs.

Full-Stack HR Management System (HRMS)

- Engineered a comprehensive, role-based web application for managing human resources.
- Implemented core modules for employee data management, a multi-level leave approval system (Employee, Manager, Admin), performance reviews, and automated payroll/payslip generation.
- Secured the application using Spring Security with distinct roles (ADMIN, HR, MANAGER, EMPLOYEE).
- **Technologies Used:** Spring Boot, Spring Security, Spring Data JPA, Thymeleaf, MySQL, HTML/CSS.

TalentScout AI (Intelligent Candidate Prioritization System)

- Engineered a scalable recruitment backend utilizing Java Spring Boot to automate candidate screening and optimize the hiring pipeline.
- Implemented a custom CandidateMaxHeap data structure, achieving $O(1)$ time complexity for retrieving top-scoring candidates based on dynamic weighted algorithms.
- Developed an AntiStarvationScheduler background service to ensure fair queue processing, and integrated "Blind Hiring" DTOs to eliminate unconscious bias by stripping Personally Identifiable Information (PII).
- **Technologies Used:** Java, Spring Boot, Custom Data Structures, RESTful APIs.

Smart IoT Mailbox

- Designed and developed a real-time hardware-software integration bridging physical delivery with digital notifications.
- Programmed ESP32 microcontrollers using C/C++ to read physical sensor data and detect mail insertion thresholds in real-time.
- Engineered an event-driven architecture to securely broadcast instant, actionable push notifications to connected devices upon mail detection.
- **Technologies Used:** C/C++, ESP32, IoT Sensors, Webhooks.

Personal Finance Manager (with Cloud Sync)

- Built a dual-database personal finance application to manage daily transactions, set category-based budgets, and track savings goals.
- Engineered a synchronization service to replicate data from a local SQLite database to a central Oracle database.
- Developed a REST API to support a vanilla JavaScript frontend and serve complex financial reports from the central database.
- **Technologies Used:** Spring Boot, Spring Data JPA, OracleDB, SQLite, Vanilla JavaScript, HTML/CSS, REST APIs.

EDUCATION

Higher National Diploma in Software Engineering

NIBM - Sri Lanka
(Continuing)

Diploma in Software Engineering

NIBM - Sri Lanka

Certificate Course in Web Design for Beginners

University of Moratuwa

Certificate Course in English Language

University of Colombo
Merit Pass

G.C.E. Advanced Level Examination

(2021) Royal College Panadura - Physical Science Stream
Combined Maths - S, Physics - S, Chemistry - S

SOFT SKILLS

- Problem-solving and logical thinking
- Teamwork and collaboration
- Time management and organization
- Adaptability and quick learning
- Attention to detail
- Critical and analytical thinking

SPOKEN LANGUAGES

- Sinhala
- English

REFERENCES

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